Guided Flow Matching for Extreme Weather Scenarios

Master's Thesis

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Date of Submission: July 30, 2026

Abstract

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Chapter 1

Introduction

Data-driven physics models have achieved state-of-the-art performance in predicting the time evolution of dynamical systems, especially in the context of high-dimensional, data-rich domains like atmospheric forecasting, and operate at a fraction of the computational cost of traditional physics-based solvers.

One particular class of data-driven physics models are generative probabilistic models. They were originally developed in the image domain, and have recently been successfully applied to the weather domain, achieving state-of-the-art performance in probabilistic forecasting. Instances of such models include GenCast, ArchesWeather, and others.

All notable applications of generative models in the image domain use guidance methods to steer the generative model towards a desired target. Instances of such applications include among others text-to-image generation, image editing, and image inpainting. In the weather domain the application of guidance use cases remains largely unexplored.

A few recent works have explored the use of guidance methods to steer generative models towards extreme weather events. ... et al generate windstorms, ... et al steer to minimise precipitation, and ... et al steer to generate heatwaves.

In this thesis we develop a general framework to specify, generate, and evaluate extreme events through the use of generative weather forecasting models. In particular, we make following contributions: 1. Firstly, we introduce a direct way to describe events of interest, taking advantage of historical data. An heuristic to define a target trajectory through the use of historical data, which can be used to define a target trajectory for the generative model to achieve, and to define a benchmark to evaluate the plausibility of the generated trajectories; 2. Secondly, we adapt the classifier free guidance framework to enable the generation of weather scenarios that achieve the specified target, while minimizing the intervention and requiring minimal tuning. A gradient based guidance method (and variants) that enables to steer forecasts towards target trajectories that achieve the guidance target while minimizes the intervention and requiring minimal tuning. 3. Finally, we develp a new evaluation benchmark to analyze the plausibility of perturbed weather trajectories, taking advantage of the same historical data used in the first step. An evaluation benchmark to test our method against plausibility tests assessing spatial, pressure-level, and cross-variable coherence.

We show that our method is able to achieve the guidance target with high reliability, and we provide a detailed analysis of the trade-offs between different design choices of the algorithm, which can be used to tailor the method to different applications.

Chapter 2

Technical Background

Generative models. Flow matching models. Existing generative weather models: GenCast, ArchesWeather, ...

Chapter 3

Problem Setting

We explicitly define a heatwave in order to design guidance methods. This means that we decide to represent a specific subset of state trajectoriesin this work. However, our method is flexible to any kind of target: you can swap your definition of the event you want to replicate and use the guidance to generate that kind of event. This really only affects the target definition.

This choice enables us to side step the problem that guiding towards a classifier's gradients or minimizing / maximizing a certain variable and stopping when we achieved the wanted result. If we specify the carachteristic of the wanted result we side step this problem entirely. The problem then shifts to retrieve a representative set of the event we have in mind. This is however a requirement for the first methodology. So we could say that the problem shifts towards defining an appropriate algorithm that can achieve a representative state similar to the specified set. Making this step explicit directly enables us to then compare the results against this set, and characterize its flaws / benefits directly.

It is really important to understand how to evaluate what we are doing here. Since we are aiming at a specific region of interest, and we are guiding only wrt the region of interest, we are loosing the global coherence of the weather states. We shall thus evaluate the ability of our method to generate locally coherent weather scenarios, over the region of interest.

In this work we focus on designing a simple guidance method and empirically test its parameters and the properties of the results generated across different settings. We put less weight on the physical realism of the definition of the guidance target and the realism of the generated trajectories, and defer this to a separate line of work. Though, we have these two in mind when designing the guidance method throughout.

Chapter 4

Guidance Design

4.1 Mask

The mask is a set of weigths (sum to 1) over a tensor having the same dimension as the weather states, non-zero at the variable and region of interest.

We experimented with a gaussian mask, but realized in our experiments that the best thing to do is equal weighting inside the region. The correct way to define the region of interest is to choose a greater mask size instead of using some weighting scheme fading out of an epicenter. The mask size should be chosen to cover the entire area of interest, ensuring that all relevant data is included in the analysis. A larger mask size can help to reduce noise and improve the accuracy of the results, while a smaller mask size may exclude important information. It is important to carefully consider the appropriate mask size for each specific application to achieve optimal results. Either way we (most probably) loose the global coherence for the guided weather states, so we might as well define directly the region of interest with a mask. Additionally, this enables us to actually make statements about "the region of interest", having equal weights everywhere. Introducing different weigths might bias the generation in a way that is not desired.

4.2 Target trajectory

In order to define plausible average trajectories of change, we define following procedure: 1. Select start state 2. Define mask 3. Retrieve historical states that are consistent with the start state and the mask (+- some tolerance) 4. Compute the distribution of average trajectories of change for the historical states 5. Filter a quantile of the distribution to define the target trajectory (e.g. 0.95 quantile for extreme events) 6. Define a gaussian distribution at each time step of the target trajectory, with mean equal to the target trajectory and variance equal to the variance of the historical states at that time step. 7. Sample enough trajectories from the gaussian distribution to define a set of target trajectories that will be used to guide the generative model. 8. Select the target trajectories of interest in terms of extremeness to build scenarios. 9. Save the trajectories and corresponding states used to define the sampling distribution. These will be used to evaluate the plausibility of the generated trajectories as means of comparison. 10. Run benchmark tests and evaluate.

This heuristic enables us to anchor the guidance target and define a benchmark at the same time. More concretly, we avoid the issue of having to tune the strength of the guidance to achieve a certain plausible result. We reverse the problem by defining the plausible result first and then designing an algorithm to achieve that result and checking that both the algorithm and its result are well behaved.

4.3 Guidance loss

The loss we use to guide the generative model is defined as follows: ...

One of the main contributions of this thesis is the realization that in the weather domain, we can define a guidance target, which can be achieved precisely. By precisely I mean that the loss can be driven to zero. While in the image domain, there is always a ground truth during training, at inference we play with different guidance strengths to identify a range where the generated images are good looking. In the weather domain however, we can define a target that can be closed by tuning the guidance strength.

This is great because instead of tuning the guidance strength to find a range of plausible results, we can define what plausible results look like a priori (guidance target sampling trajectory defined through historical data and plausibility tests), and optimize the guidance strength to achieve this target.

We essentially have a very similar setting to the guided diffusion for precipitation paper, but we make it more general in these ways:

- we define a guidance target explictly, in contrast to observe results across different lambdas
- we experiment with different a_t trajectories, in addition to guiding computing the gradient at the first step

- we test level of intensity by shifting the intenisty parameter from the flow (lambda) to the trajectory definition (phi_n)
- Our loss is driven to zero in terms of the difference from the target, instead of being minimized iteratively (a more explicit way to encode the target scenario)

4.4 Guidance method

Assumption about using the gradient. We explicitly assume that in order to produce realistic result we have to tie the perturbations to the models' gradient. We use the none-gradient tied perturbations as baseline for all our experiments.

The hope is also that the gradient is informative in the sense that: if a method pushes a not secondary target variable to a certain distribution / average value, we would also expect that if we push this variable to the achieved average value, that the previously targeted variable would achieve the same result. This we explicitly test in test5.

The base guidance algorithm we use works as follows: ...

4.5 Guidance algorithms

4.5.1 CLOSE-GAP

Greedy

Step is normalized to close the gap as prescribed by the guidance schedule.

4.5.2 OPTIMIZE-GAIN

Secant method

The step is allowed to take the gradient norm into consideration. Since the loss is almost linear with respect to the guidance strength, we can use a secant method to find the guidance strength that minimizes the loss.

In this case we define the secant method as follows:

4.5.3 OPTIMIZE-SCHEDULE

The step is allowed to take the gradient norm into consideration. The schedule is optimized to minimize the loss while also complying to a regularization term. Possible regularization terms include the total guidance. We can dream up arbitrary penalty terms such as angular defelction, ...

The problem of this optimization is that the gradient applyed for the guidance at each step changes with respect to the schedule. The optimization is therefore highly non-convex and very hard to solve. In fact, the optimizer will converge very fast to a local minimum, highly influenced by the initialization of the schedule.

LBFGS

Chapter 5

Related Work

Appendices

Appendix A

Appendix

Eidesstattliche Erklärung

Der/Die Verfasser/in erklärt an Eides statt, dass er/sie die vorliegende Arbeit selbständig, ohne fremde Hilfe und ohne Benutzung anderer als die angegebenen Hilfsmittel angefertigt hat. Die aus fremden Quellen (einschliesslich elektronischer Quellen) direkt oder indirekt übernommenen Gedanken sind ausnahmslos als solche kenntlich gemacht. Die Arbeit ist in gleicher oder ähnlicher Form oder auszugsweise im Rahmen einer anderen Prüfung noch nicht vorgelegt worden.

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